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A comprehensive review of efficient deep learning for clinical medical imaging deployment

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Abstract

Deep learning has revolutionized medical image analysis, playing a vital role in modern clinical applications. However, the deployment of large-scale models in real-world clinical settings remains challenging due to high computational costs, latency constraints, and patient data privacy concerns associated with cloud-based processing. To address these bottlenecks, this review provides a comprehensive synthesis of efficient and lightweight deep learning architectures specifically tailored for the medical domain. We categorize the landscape of modern efficient models into three primary streams: Convolutional Neural Networks (CNNs) for local inductive bias, Lightweight Transformers for global context under constrained attention, and emerging Linear Complexity Models for scalable global context. Furthermore, we examine key model compression strategies (including pruning, quantization, knowledge distillation, and low-rank factorization) and evaluate their efficacy in maintaining diagnostic performance while reducing hardware requirements. By identifying current limitations and discussing the transition toward on-device intelligence, this review serves as a roadmap for researchers and practitioners aiming to bridge the gap between high-performance AI and resource-constrained clinical environments.

Keywords Medical image analysis, Lightweight deep learning, Edge AI, State space models, Model compression, Point-of-care

1 Introduction

The convergence of Artificial Intelligence (AI) and healthcare has fundamentally reshaped the landscape of modern medicine, driven by the availability of high-dimensional medical imaging modalities and increasingly sophisticated learning algorithms. From radiology and pathology to ophthalmology and cardiology, deep learning systems have demonstrated expert-level performance across a wide range of diagnostic tasks, often surpassing human benchmarks under controlled experimental conditions [1, 2]. These advances have fueled optimism regarding the transformative potential of AI in

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improving diagnostic accuracy, streamlining clinical workflows, and alleviating workforce shortages.

However, the dominant paradigm underpinning many of these successes has been the relentless scaling of model size, depth, and computational complexity. Inspired by breakthroughs in natural image analysis and large-scale foundation models, medical imaging research has increasingly embraced architectures with hundreds of millions, or even billions of parameters. While such scaling often yields incremental gains in benchmark performance, it comes at a rapidly escalating computational cost. This trend has reached a critical inflection point, where marginal improvements in accuracy are outweighed by substantial increases in training time, memory footprint, inference latency, and energy consumption.

The operational implications of this imbalance are profound. Large-scale medical imaging models demand specialized hardware accelerators, continuous maintenance, and significant energy resources, leading to prohibitive deployment costs for many healthcare systems. Beyond economic considerations, these models raise growing concerns regarding environmental sustainability. Recent analyses indicate that the carbon footprint associated with training and deploying state-of-the-art deep learning models can be substantial, with emissions comparable to those generated by multiple automobiles over their lifetime [3, 4]. It is important to note that these estimates predominantly capture operational energy use during training and inference and do not, in general, account for the embodied carbon associated with the manufacture of the underlying GPUs, accelerators and edge devices, which lifecycle assessments suggest can constitute a comparable share of the total environmental cost. Such findings have catalyzed the emergence of the "Green AI" movement, which advocates for efficiency-aware model development as a core research objective rather than an afterthought [5]. From this lifecycle perspective, efficient on-device architectures contribute to sustainability not only by lowering operational energy demand, but also by reducing the impetus to upgrade to newer accelerators, thereby extending hardware lifetimes and amortizing embodied emissions over longer deployment horizons. In the context of healthcare, where ethical responsibility extends beyond individual patients to population-level impact, this shift is particularly salient.

Crucially, high algorithmic performance does not automatically translate into clinical utility. Despite impressive results in retrospective studies, real-world integration of AI systems into clinical workflows remains limited. This gap reflects a fundamental performance paradox (the deployment paradox), in which models optimized for accuracy under idealized conditions fail to meet the practical constraints of clinical environments. These constraints can be conceptualized as a persistent "trilemma" encompassing data privacy, latency requirements, and infrastructure inequality.

First, stringent privacy regulations and ethical considerations severely restrict cloud-based processing of sensitive medical data. Contrary to long-standing assumptions, medical images such as MRI scans are not inherently anonymous. Strikingly, recent studies demonstrate that facial reconstruction and recognition techniques can re-identify MRI participants with accuracies reaching up to 83%, exposing patients to significant re-identification risks [6, 7]. As a result, transmitting raw imaging data to centralized cloud servers is increasingly viewed as incompatible with modern data protection frameworks, including the General Data Protection Regulation (GDPR) and the Health Insurance

Portability and Accountability Act of 1996 (HIPAA). These concerns necessitate a shift toward on-device or edge-based inference, where data remain within institutional or point-of-care boundaries [8].

Second, latency constraints impose strict real-time requirements in many clinical scenarios. In applications such as robot-assisted surgery, interventional imaging, and Point-of-Care Ultrasound (POCUS), inference delays of even a few hundred milliseconds can degrade clinical performance or compromise patient safety [9, 10]. While cloud offloading is computationally attractive, it cannot reliably guarantee low-latency responses due to network congestion, bandwidth variability, and connectivity disruptions. This makes lightweight, locally executable models a prerequisite rather than a design preference.

Third, infrastructure inequality poses a major barrier to the equitable deployment of AI-driven diagnostics. Many rural hospitals and healthcare facilities in low-income and middle-income countries lack access to high-end GPUs, stable electricity, or high-bandwidth internet connections. As a result, computationally intensive models exacerbate existing disparities in global health by disproportionately benefiting well-resourced institutions [11]. Addressing this imbalance requires AI systems that are robust, efficient and deployable across heterogeneous hardware platforms, including low-power edge devices [12]. Unlike consumer vision domains such as photo tagging or recommendation systems, where cloud offloading and best-effort latency are routinely acceptable, the medical setting tightens all three constraints simultaneously, making efficiency a prerequisite for clinical deployment rather than a performance optimization.

Collectively, these challenges necessitate a fundamental paradigm shift in medical AI research. As illustrated in Fig. 1, the field is moving away from a reliance on resource-intensive, cloud-centric models toward a "deployment-first" framework. This perspective contrasts the limitations of the traditional approach, such as privacy risks and latency bottlenecks, with the improvements offered by efficient, edge-native designs. Consequently, efficiency is reframed not merely as a secondary optimization step, but as a central criterion that determines whether AI systems can bridge the gap between algorithmic potential and real-world clinical value.

Motivated by this perspective, this review synthesizes the rapidly evolving landscape of efficient deep learning for medical imaging. We provide a structured roadmap that spans both architectural innovation and model optimization strategies aimed at bridging the gap between algorithmic performance and clinical deployment. Specifically, we categorize architectural advances into three major streams: (1) efficient convolutional neural networks (CNNs), which leverage inductive biases and parameter sharing for resource-aware learning; (2) lightweight Transformer-based models that seek to mitigate the quadratic complexity of self-attention; and (3) emerging linear-complexity architectures, including Mamba and state space models, which promise scalable global context modeling for high-resolution medical images. Complementing this architectural analysis, we examine key model compression techniques, such as network pruning, quantization, knowledge distillation, and low-rank factorization. These techniques further reduce computational overhead while preserving diagnostic fidelity within clinically acceptable margins. By critically analyzing the trade-offs among model size, energy consumption, latency, and clinical sensitivity, this review aims to guide researchers and practitioners toward the development of AI systems that are not only high-performing, but also truly deployable in resource-constrained clinical environments.

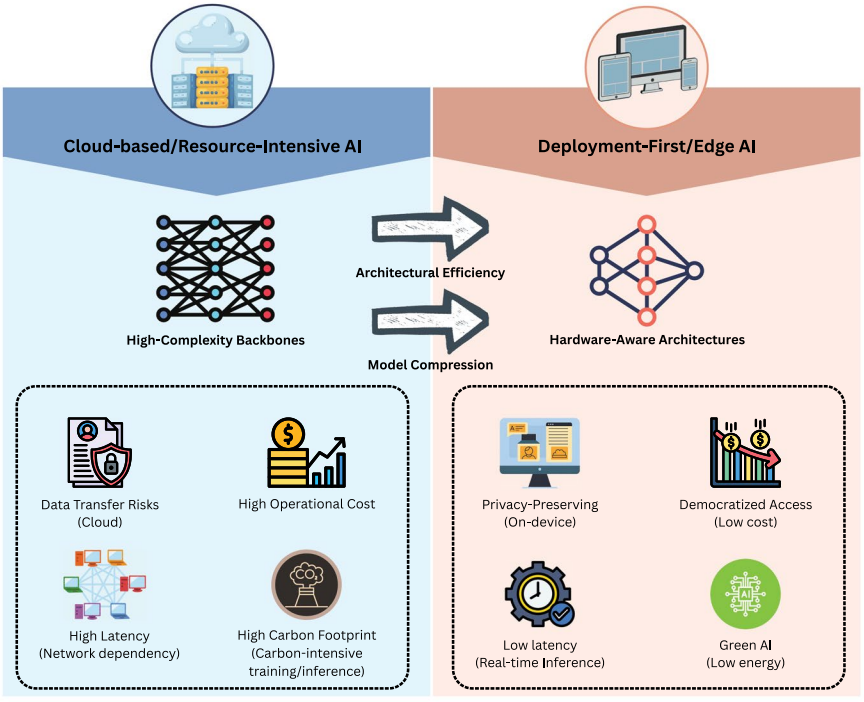


Fig. 1 The Paradigm Shift toward Deployment-First Medical AI. The figure contrasts the legacy *Cloud-based/Resource-Intensive AI* paradigm (left panel, blue) with the emerging *Deployment-First/Edge AI* paradigm (right panel, peach). The left panel summarizes four limitations of centralized inference: data-transfer privacy risks, high operational cost, network-dependent latency, and a carbon-intensive training and inference footprint. The right panel summarizes the corresponding improvements achievable through on-device deployment: privacy preservation (data remain on-device), democratized access (low-cost hardware), low-latency real-time inference, and reduced operational energy (Green AI). The two horizontal arrows linking the panels denote the two technical levers driving this shift: *Architectural Efficiency* (top arrow), referring to hardware-aware backbones such as lightweight CNNs, hybrid Transformers, and linear-complexity state space models (§4); and *Model Compression* (bottom arrow), referring to pruning, quantization, knowledge distillation, and low-rank factorization (§5). Together, these levers transition high-complexity backbones into hardware-aware architectures suitable for clinical edge deployment

2 Research methodology

This article is presented as a *narrative, deployment-oriented review* rather than a PRISMA-style systematic review. Its objective is to synthesize, taxonomize, and critically discuss efficient deep learning architectures and model compression techniques for clinical medical imaging deployment, rather than to estimate effect sizes across a closed body of studies. Accordingly, we adopt a structured but non-exhaustive literature selection protocol, the parameters of which are disclosed below to support transparency and reproducibility. For clarity, throughout this review the term "AI" refers to deep-learning–based methods unless otherwise indicated; broader senses (e.g., healthcare AI policy, AI regulation) are noted in context.

2.1 Scope and guiding questions

The review is organized around three guiding questions that mirror the structure of the manuscript:

1. Which architectural paradigms (efficient CNNs, lightweight Transformers and hybrids, and linear-complexity State Space Models) have been adapted to medical imaging, and how do their design choices interact with clinical deployment constraints?

2. How do model compression techniques (pruning, quantization, knowledge distillation, and low-rank factorization) translate into clinically acceptable efficiency–fidelity trade-offs?
3. Which evaluation practices and reporting conventions are required to make efficiency claims interpretable across heterogeneous edge environments?

2.2 Information sources and search strategy

We searched for five sources: IEEE Xplore, PubMed, Scopus, Google Scholar, and arXiv. The primary search window was January 2017 to December 2025, chosen to span from the emergence of mobile-oriented backbones (MobileNet V1, ShuffleNet, ESPNet) through the recent wave of Mamba-based architectures, with the strongest emphasis on the 2022–2025 period. Seminal works before 2017 (e.g., VGG, ResNet) were included where required to provide architectural context. To capture the heterogeneity of the field, we used three complementary query groups:

- **Group 1 – Lightweight CNN and Transformer Architectures.**

```
("Medical Image Analysis" OR "Medical Imaging") AND
("Lightweight" OR "Efficient" OR "Compact" OR "Mobile") AND
("Deep Learning" OR "CNN" OR "Transformer" OR "ViT") AND
("Real-time" OR "On-device" OR "Edge" OR "Point-of-care")
```

- **Group 2 – Linear Complexity Models (State Space Models).**

```
("Medical Imaging" OR "Medical Image Segmentation" OR
"Medical Image Classification") AND ("Mamba" OR "State
Space Model" OR "SSM" OR "Vision Mamba" OR "Selective
Scan")
```

- **Group 3 – Model Compression and Deployment.**

```
("Medical Imaging") AND ("Knowledge Distillation" OR
"Quantization" OR "Pruning" OR "Low-rank Factorization"
OR "Teacher-Student") AND ("Deployment" OR "Edge" OR
"Embedded" OR "Mobile" OR "FPGA" OR "NPU")
```

In addition to keyword searches, we performed backward and forward citation tracing on highly cited works in each group (e.g., MobileNet, MobileViT, VMamba, LightM-UNet) to capture closely related studies that the keyword queries might have missed. Inference toolchains and compiler frameworks (TensorRT, ONNX Runtime, OpenVINO, CoreML, TVM) were retrieved directly from their official documentation as primary sources.

2.3 Eligibility criteria

Inclusion criteria.

1. **Domain relevance.** Studies addressing medical imaging across 2D modalities (X-ray, ultrasound, dermoscopy, fundus, endoscopy), 2D slices and 3D volumes of CT and MRI, and gigapixel whole-slide pathology images.
2. **Task relevance.** Diagnostic tasks central to clinical workflows: classification, segmentation, and detection. Generative and reconstruction tasks are discussed only when directly relevant to deployment efficiency (e.g., distilled denoisers in §5).
3. **Efficiency reporting or claim.** The work either reports at least one efficiency-relevant metric (parameter count, FLOPs, latency or FPS, peak memory, or energy per inference) alongside standard accuracy or Dice measures, or proposes an architectural or compression technique whose efficiency implications can be assessed from its design.
4. **Contribution type.** Original research proposing or empirically evaluating efficient architectures, model compression methods, or deployment-oriented evaluation frameworks; high-impact survey papers used for situating the present review (§3).**Exclusion criteria.**

1. Studies focused exclusively on generative modeling, image reconstruction, or self-supervised pretraining without an efficiency or deployment component.
2. Studies in which neither computational complexity (parameters/FLOPs) nor wall-clock efficiency (latency/FPS/energy) can be inferred or estimated.
3. Non-peer-reviewed preprints unless they introduce widely-adopted architectures (e.g., Mamba, MedMamba) or are necessary to reflect the current state of the rapidly evolving SSM literature.
4. Studies addressing non-imaging clinical modalities (e.g., text, tabular EHR, signal-only ECG) that fall outside the scope of medical image analysis.

2.4 Selection and synthesis process

The extracted records were selected by title and abstract for relevance, followed by a full-text review of the candidate studies. Studies satisfying the inclusion criteria were then grouped into the taxonomy used throughout the manuscript: (i) efficient CNNs, (ii) lightweight Transformers and hybrid CNN–Transformer architectures, (iii) linear-complexity State Space Models, and (iv) model compression techniques. Studies addressing cross-cutting themes such as the compiler gap, safety-critical optimization, and equity were aggregated into the discussion (§6).

Because this is a narrative review rather than a meta-analysis, we do not report pooled effect sizes or PRISMA flow counts. Quantitative comparisons in Tables 5 and 6 are reported as published by the original authors, with explicit caveats regarding the influence of hardware, runtime, precision, and input resolution on the comparability of latency and FPS measurements.

2.5 Limitations of the review protocol

We acknowledge three limitations of this protocol. First, the search is non-exhaustive: novel architectures and compression techniques appear on arXiv on a near-weekly basis, and any narrative review necessarily reflects a snapshot in time. Second, the SSM and Mamba-based literature is recent and rapidly evolving; conclusions drawn for this paradigm should be interpreted as emerging rather than settled. Third, the deployment-first focus intentionally constrains the scope: studies that are clinically important but do not

take into account efficiency considerations are not covered, and readers seeking accuracy-only benchmarks should consult complementary surveys cited in §3.

3 Related work

The rapid adoption of deep learning in medical imaging has led to a substantial body of survey literature. Existing reviews provide valuable perspectives on architectural innovation and learning paradigms, yet they are predominantly organized around specific model families or algorithmic advances. As a result, questions concerning deployment readiness under clinical resource constraints are often addressed only implicitly. This section situates the present survey within this evolving literature.

3.1 Architecture-centric surveys in medical imaging

A large fraction of existing reviews focuses on individual architectural families. Convolutional neural networks remain the most extensively studied paradigm in medical image analysis, with multiple surveys documenting their widespread adoption across diagnostic tasks and the evolution of convolutional architectures for classification and segmentation [13–20]. These studies primarily emphasize improvements in diagnostic accuracy and architectural design, while deployment-related constraints such as inference latency, memory footprint, and energy consumption receive comparatively limited attention.

The emergence of vision transformers has motivated a parallel body of survey literature, examining their capacity to capture long-range dependencies in medical images and their adaptation to various imaging tasks [21]. Subsequent surveys further analyze transformer variants tailored for medical imaging applications, often acknowledging the substantial computational cost of self-attention mechanisms. However, efficiency considerations are typically discussed within the transformer family itself, and direct comparisons with optimized convolutional architectures or alternative sequence modeling approaches under identical deployment conditions remain uncommon.

More recently, state space models have been proposed as computationally efficient alternatives to transformers. Dedicated surveys review the application of architectures such as Mamba across medical imaging tasks including classification, segmentation, and image restoration, highlighting favorable theoretical scaling properties and architectural design choices that mitigate quadratic complexity [22–24]. Nevertheless, empirical analyses of real-world deployment behavior, including hardware-specific latency, memory usage, and energy efficiency, are still relatively limited.

In addition to general architectural surveys, several focused reviews examine efficiency and compression within specific model families. Surveys on vision transformer compression for edge deployment provide taxonomies of pruning, quantization, and distillation strategies alongside discussions of hardware-aware optimization, while focused analyses of state space models explore design variants with attention to computational complexity and runtime behavior. Although these works offer detailed technical insights within individual paradigms, they do not explicitly frame efficiency trade-offs in relation to clinical deployment constraints such as privacy preservation, latency requirements, and infrastructure heterogeneity.

Collectively, architecture-centric surveys offer in-depth perspectives within individual model families but provide limited guidance on cross-architectural trade-offs that are critical for deployment in clinical environments.

3.2 Efficiency-aware and deployment-oriented reviews

Beyond architecture-centric analyses, a smaller body of literature explicitly addresses efficiency and deployability in medical artificial intelligence. Surveys on edge deep learning and lightweight models discuss compression techniques, hardware constraints, and runtime considerations for point-of-care and mobile devices, emphasizing latency, memory usage, and power consumption [25–27]. However, these evaluations often span broad application domains and rarely establish a unified framework for comparing fundamentally different architectural paradigms under consistent clinical constraints.

In digital pathology, where whole-slide images frequently exceed gigapixel resolution, efficiency challenges are commonly addressed at the pipeline level rather than solely through backbone design. Approaches based on ultra-high-resolution tokenization, feature aggregation, and multiple-instance learning aim to reduce computational and memory requirements prior to or during inference. Recent methods demonstrate how scalable whole-slide image analysis can be achieved without exhaustively processing all patches, enabling global context modeling at substantially lower computational cost in clinically relevant high-resolution settings [28, 29].

Complementary advances in multiple-instance learning further illustrate deployment-conscious design beyond backbone efficiency. Prototype-based Multiple Instance Learning methods and graph-based coarsening frameworks seek to improve training and inference efficiency while preserving interpretability at the instance or region level, an important consideration in pathology workflows where clinical trust and localized evidence are essential. These approaches operate orthogonally to backbone selection and model compression strategies, underscoring that deployment considerations extend beyond architectural efficiency alone.

Another important aspect of deployment readiness concerns inference toolchains and compiler support. Practical efficiency gains often depend on execution frameworks such as TensorRT, ONNX Runtime, OpenVINO, CoreML, NNAPI, and TVM, which translate trained models into optimized hardware-specific kernels [30–35]. Architectural efficiency does not automatically translate into real-world speedups unless supported by the underlying runtime and compiler ecosystem, and techniques such as structured sparsity, operator fusion, and mixed-precision inference yield measurable benefits only when explicitly supported by the deployment backend. This factor is rarely considered in prior survey literature.

Overall, existing surveys provide valuable insights into individual architectural families, efficiency techniques, and domain-specific deployment strategies, yet the literature remains fragmented with limited synthesis across architectures from a unified deployment-oriented perspective. Against this backdrop, the present review makes four explicit contributions that, to our knowledge, are not jointly addressed by existing surveys: (i) a unified cross-paradigm taxonomy that compares efficient CNNs, lightweight Transformers and hybrids, and linear-complexity State Space Models under a single deployment-oriented lens, rather than within isolated model families; (ii) elevation of the compiler and runtime gap to a first-class analytical concern, distinguishing theoretical efficiency (FLOPs, parameter count) from realized latency and energy on heterogeneous edge hardware; (iii) a deployment-first reporting checklist (Table 4) that operationalizes the disclosure of hardware–software stacks, inference protocols, and reliability metrics required for reproducible efficiency claims; and (iv) an explicit safety-aware framing of

model compression, in which calibration, rare-class sensitivity, and failure-mode analysis are treated as core evaluation criteria rather than optional reporting items. Rather than revisiting algorithmic details or proposing new models, this review organizes prior work according to how different architectural design choices address practical clinical constraints, including latency, energy consumption, privacy preservation, and hardware accessibility. This perspective motivates the taxonomy of deployment-ready architectures presented in the following section.

4 Evolution of deployment-ready architectures: from inductive bias to linear global context

The landscape of deployment-ready architectures has evolved through three major technological waves, as illustrated in Fig. 2. This evolution reflects a continuous pursuit of optimal trade-offs between diagnostic fidelity and computational cost, moving from the rigid local features of CNNs to the dynamic global context of Transformers, and most recently to the linear-complexity modeling of State Space Models.

Before dissecting specific architectural families, however, we must address a pervasive misconception in model evaluation: treating Floating-point Operations (FLOPs) and parameter counts as sole proxies for efficiency. Historically, researchers equated low FLOP counts with deployment readiness, assuming an idealized computing environment where arithmetic intensity dominates performance. Real-world clinical deployment tells a different story, particularly on edge devices such as mobile NPUs or embedded GPUs, where latency depends less on calculation volume and more on Memory Access Cost (MAC) and degree of parallelism [36]. This discrepancy creates an efficiency gap: architectures optimized purely for FLOPs often suffer from low hardware utilization through

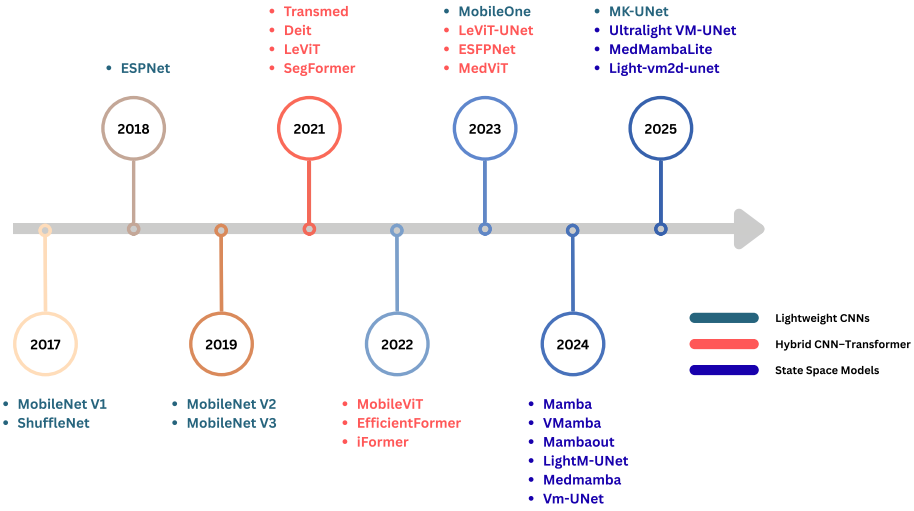


Fig. 2 Chronological evolution of efficient deep learning architectures in medical imaging (2017–2025). The timeline shows the transition across three architectural waves, each annotated with its primary deployment characteristic: *Lightweight CNNs* (teal, local-feature focus and mature edge toolchain support), spanning MobileNet V1/V2/V3, ShuffleNet, ESPNet, MobileOne, MK-UNet, and related designs; *Hybrid CNN-Transformer* architectures (red, bridging local efficiency with global context), including TransMed, DeiT, LeViT, SegFormer, MobileViT, EfficientFormer, iFormer, LeViT-UNet, ESFPNet, and MedViT; and *State Space Models* (blue, linear-complexity global context for high-resolution and volumetric data), comprising Mamba, VMamba, MambaOut, LightM-UNet, MedMamba, VM-UNet, Ultralight VM-UNet, MedMambaLite, and Light-VM2D-UNet. The progression reflects a continuous shift in the dominant inductive bias from rigid locality to dynamic global modeling, with each wave progressively addressing the compute-memory-context trade-off central to clinical deployment

excessive use of fragmented depth-wise convolutions or sparse connections, resulting in inference times that contradict their theoretical lightness. Consequently, the modern definition of deployment readiness has shifted from FLOP-centric minimization to system-level optimization, prioritizing characteristics such as memory contiguity, operator fusion compatibility, and dimension consistency. We adopt this hardware-aware perspective as the analytical lens for reviewing the three architectural paradigms in the following.

4.1 CNNs as the mature standard for local efficiency

Early successes of deep learning in medical imaging relied on architectures like VGG and ResNet, which prioritized representational capacity over computational feasibility [37, 38]. While effective in retrospective studies utilizing high-end GPUs, these dense, heavy models proved ill-suited for the emerging paradigm of point-of-care and portable diagnostics. To bridge this gap, research focus shifted toward rethinking the fundamental building blocks of CNNs, leading to adoption of Depth-wise Separable Convolutions (pioneered by the MobileNet family [39–41]) and Channel Shuffling (ShuffleNet [42]). By decoupling spatial filtering from channel mixing, these architectures reduced theoretical computational cost by an order of magnitude, establishing the first standard for lightweight medical analysis. However, widespread deployment revealed a critical discrepancy: theoretical efficiency did not guarantee real-world speed, as inference latency on edge devices was often bottlenecked by fragmented memory access rather than arithmetic intensity. Depth-wise convolutions, despite being FLOP-efficient, can suffer from suboptimal bandwidth utilization on certain hardware. Addressing this, recent latency-aware designs like MobileOne [43] introduce Structural Re-parameterization, a technique that decouples training-time complexity (multi-branch topology for convergence) from inference-time execution (single-path topology for speed), enabling sub-millisecond latency on mobile devices by minimizing MAC and maximizing operator fusion.

Despite these advances, generic lightweight backbones often struggle to capture subtle, irregular structures characteristic of medical images, as fixed, localized receptive fields cannot adequately represent diffuse lesion boundaries or fine vascular networks. This limitation is intrinsic to the convolutional inductive bias: translation-equivariant local kernels efficiently capture sharp, recurring textures but underweight long-range structural cues such as organ-level symmetry or spatially distributed pathologies, motivating the hybrid and global designs discussed in §4.2 and §4.3. It has also motivated a wave of task-specific micro-architectures tailored for medical constraints. MK-UNet [44] and LDMRes-Net [45] employ multi-kernel depth-wise convolutions to aggregate features at multiple scales simultaneously without the heavy overhead of traditional pyramids, allowing precise boundary delineation in segmentation tasks with extreme parameter efficiency (MK-UNet uses only 0.027M parameters). In a similar spirit, EGE-UNet [46] achieves state-of-the-art skin-lesion segmentation with only 50 KB of parameters by combining a Group multi-axis Hadamard Product Attention module with a Group Aggregation Bridge, demonstrating that competitive segmentation accuracy is achievable at the smallest parameter scales encountered in the literature. Similarly, U-Lite [47] leverages Axial Convolutions to expand the receptive field along distinct spatial axes, essential for organ-level context, while remaining $35\times$ smaller than a standard U-Net. In interventional scenarios such as endoscopy and ultrasound, where latency supersedes

peak accuracy and maintaining high throughput is non-negotiable, ESPNet [48] exemplifies the solution by utilizing Efficient Spatial Pyramids with dilated convolutions. This approach expands the receptive field to capture global context without the resolution loss associated with pooling, enabling segmentation at over 100 FPS on edge GPUs, a critical requirement for closed-loop visual feedback.

Collectively, the literature reviewed above establish CNNs as the most mature paradigm for local efficiency, demonstrating through evolution from generic depth-wise separability to hardware-aware re-parameterization and medical-specific micro-designs that diagnostic fidelity can coexist with stringent resource constraints. However, their reliance on localized operators imposes intrinsic limitations in modeling long-range dependencies, setting the stage for the hybrid and attention-based paradigms discussed next.

4.2 The hybrid compromise: reconciling local efficiency with global context

Although CNNs have proven robust for local feature extraction, their reliance on finite receptive fields imposes a fundamental limitation in modeling long-range dependencies, a constraint particularly consequential in medical imaging where diagnostic reasoning often depends on spatial relationships between distant anatomical structures (consider identifying global symmetry in brain MRI or contextualizing diffuse pathologies like interstitial lung disease). Vision Transformers (ViTs) theoretically resolve this by capturing global interactions via self-attention [49], yet their quadratic complexity ($O(N^2)$) and lack of inductive bias render them computationally prohibitive for resource-constrained edge devices. To bridge this gap, Hybrid Architectures have emerged as a pragmatic compromise, strategically embedding attention mechanisms within convolutional pipelines to amortize the cost of global context modeling across efficient local representations. The evolution of hybrid designs reflects a critical shift from simple architectural integration to hardware-aware co-design: early approaches such as MobileViT [50] treated Transformers as drop-in modules, unfolding feature maps into non-overlapping patches to capture global context before refolding them for spatial processing, but recent deployment analysis reveal that these frequent reshaping operations create significant compiler friction on mobile NPUs, as switching between 4D-tensor and 3D-sequence representations often bottlenecks actual inference latency. This bottleneck is highly platform-dependent: on server-grade GPUs, fine-grained patch tokenization is largely absorbed by high-bandwidth memory and parallel kernel execution, whereas on mobile NPUs and GPUs, the same unfold, permute, and reshape operations require explicit memory-layout transformations whose latency cost grows with token count and dominates inference time on high-resolution medical images, where doubling spatial resolution increases the number of tokens fourfold and the cost of self-attention sixteenfold.

Addressing this, next-generation hybrids like EfficientFormerV2 [51] prioritize dimension-consistent design by retaining a 4D convolutional structure for the majority of the network and restricting computationally expensive attention mechanisms to the final stages, minimizing reshaping overhead and allowing effective leverage of mobile compilers (CoreML, SNPE) to achieve latencies comparable to MobileNet while retaining the global reasoning of Transformers. Similarly, iFormer [52] tackles the unique frequency characteristics of medical images by employing an Inception-like mixer that captures high-frequency information (local textures, edges) and low-frequency information

(global shape, structure) in parallel paths, optimizing capture of fine lesion boundaries without the latency penalty of dense self-attention. Beyond classification, hybrid architectures have proven particularly adept at dense prediction tasks essential for clinical workflows: in volumetric segmentation where fixed input sizes are rarely guaranteed, SegFormer [53] eliminates rigid positional encodings in favor of a hierarchical Mix-Transformer encoder, allowing robust generalization across varying image resolutions and slice thicknesses, while architectures like LeViT-UNet [54] further optimize speed by aggressively integrating convolutions and Batch Normalization into the Transformer encoder, reducing memory access overhead to enable segmentation speeds of up to 114 FPS and validating that attention mechanisms, when carefully constrained, are compatible with latency-critical interventional applications such as real-time endoscopy analysis [55].

Finally, the hybrid paradigm offers distinct advantages in data efficiency and multimodal reasoning, addressing the scarcity of annotated medical data: through token-based distillation strategies such as those in DeiT [56], hybrids inherit the strong inductive biases of CNN teachers, enabling faster convergence on small datasets where pure Transformers often struggle, while frameworks like TransMed [57] leverage the attention mechanism to model cross-modal dependencies (such as the correlation between T1-weighted and T2-weighted MRI sequences), outperforming single-modality baselines that lack the capacity to fuse heterogeneous feature spaces.

In summary, hybrid architectures successfully balance the trade-off between local efficiency and global context, though their reliance on attention mechanisms, even when optimized, still incurs non-trivial memory access costs for high-resolution or long-sequence inputs, motivating exploration of linear-complexity paradigms such as State Space Models discussed in the next section.

4.3 The linear revolution: state space models and mamba

Although hybrid CNN–Transformer architectures substantially reduce the practical cost of global context modeling, they do not fundamentally resolve the quadratic scaling of self-attention: as image resolution increases, attention-based models still incur rapidly growing memory and latency overheads, limiting their applicability to gigapixel histopathology slides, long cine MRI sequences, or high-resolution 3D volumetric scans. This bottleneck has motivated a search for architectures that can preserve global receptive fields while eliminating quadratic complexity altogether, culminating in the decisive shift to State Space Models (SSMs) for deep learning, particularly the Mamba architecture [58]. Originally developed for long-range sequence modeling, Mamba offers a compelling proposition for medical imaging: global information propagation with strictly linear computational complexity ($O(N)$), conceptually bridging the gap between recurrent models and Transformers by replacing explicit attention with selective state updates that enable long-context reasoning without pairwise token interactions. This property positions SSMs as a fundamentally different and potentially more scalable solution to the deployment challenges faced by attention-based models, though adapting Mamba to medical images is non-trivial, as the original formulation operates on 1D sequences whereas medical data are inherently 2D or 3D and exhibit strong spatial locality. The choice of scan strategy is itself a non-trivial design axis with direct implications for anatomical coherence. Naive raster scans flatten the image row by row, so tokens that are

spatially adjacent across rows (e.g., pixel (i, j) and $(i+1, j)$) become arbitrarily distant in the 1D sequence, breaking the local continuity that medical structures such as vessel walls, cortical surfaces, or tumor boundaries depend on [59, 60]. Bi-directional and four-directional scans such as VMamba's CSM partially mitigate this causality mismatch by aggregating context from multiple traversal orders, at the cost of three to four redundant scans per layer. Locality-preserving alternatives based on space-filling curves, such as Hilbert and Peano curves, map spatially adjacent 2D coordinates to nearby positions in the 1D sequence and can in principle preserve fine-detail continuity more faithfully than raster traversal. However, these curves are more expensive to compute, lack native operator support in mobile inference frameworks, and have not yet been broadly validated for 3D volumetric medical data, where the 2D→1D collapse becomes a 3D→1D collapse and locality loss is correspondingly amplified. In practice, cross-scan currently offers the most stable empirical compromise between contextual coverage, memory access efficiency, and toolchain maturity, while learnable or content-adaptive scan strategies represent an active area of investigation in medical SSMs [61]. Despite their promise, SSMs are not universally superior to attention-based models, as critical analysis (MambaOut) shows that the advantages of SSMs are highly task-dependent [62]: for dense prediction problems such as segmentation, where accurate delineation of long-range boundaries is essential, SSMs consistently outperform purely convolutional baselines, while for image-level classification tasks with limited spatial complexity, gated CNNs or lightweight hybrids may achieve comparable performance with lower architectural complexity. This distinction highlights an important theme: efficiency gains from SSMs stem not only from novelty, but also from alignment between architectural inductive bias and task structure. Based on currently reported results, the most striking impact of Mamba-based models appears in resource-constrained segmentation scenarios. An early example is U-Mamba [63], which embeds Mamba blocks into a CNN-based U-Net encoder under a self-configuring nnU-Net pipeline and reports improved long-range dependency modeling across 2D and 3D abdominal organ segmentation, organ-at-risk delineation, and endoscopy instrument segmentation, establishing the CNN–SSM hybrid template, subsequently extended by lighter-weight variants. LightM-UNet integrates Mamba blocks into a U-Net-style architecture to achieve state-of-the-art segmentation accuracy with approximately 1 M parameters [64], while Ultralight VM-UNet pushes efficiency to an extreme by employing a Parallel Vision Mamba (PVM) layer to deliver competitive performance with only 0.049M parameters and 0.06 GFLOPs [65], representing an extreme point in the current efficiency landscape for deployable medical segmentation models and highlighting results that would be unattainable under quadratic attention scaling.

Beyond parameter efficiency, recent work explores biologically inspired scanning strategies to improve feature retention: Light-VM2D-UNet introduces Wavefront Scanning, which propagates information in a manner analogous to physical signal diffusion through tissue [66], preserving anatomical continuity more effectively compared to raster or cross-scan traversal and suggesting that scanning order is not merely a computational artifact but a meaningful design choice in medical SSMs. For classification tasks, hybridization remains beneficial, as MedMamba interleaves convolutional layers with SSM blocks to combine the strong inductive bias of CNNs with the long-range modeling capability of Mamba [67], while this design philosophy extends naturally to deployment-oriented optimization through MedMambaLite, which further distills

and re-parameterizes the architecture for edge accelerators to achieve a reported 63% improvement in energy efficiency on platforms such as NVIDIA Jetson Orin Nano compared to its non-optimized teacher model [68]. This result emphasizes that the value of SSMs lies not only in speed or parameter count, but also in enabling sustainable, battery-powered medical AI systems aligned with Green AI objectives.

Despite these advances, a critical evaluation of SSMs reveals three categories of limitation that warrant caution before broad clinical adoption. First, training stability is not as benign as the linear-complexity framing suggests: recent theoretical and empirical analysis show that Mamba's input-dependent selection mechanism produces gradient magnitudes that scale unfavorably with sequence length, leading to unstable optimization on long-range benchmarks and motivating modifications such as block-wise selective dynamics and reduced learning rates on the selection parameters [69]. In dense prediction tasks with high-resolution inputs, where the sequence length becomes large after patchification, this instability is most pronounced in deep variants and can erode the reproducibility of the reported results. Second, performance remains sensitive to the chosen scan order, as discussed above; in contrast to attention mechanisms which are inherently order-equivariant, SSMs require an explicit traversal that hard-codes a directional inductive bias not always aligned with anatomy. Third, the software ecosystem is markedly less mature than that of CNNs and Transformers: the selective scan operator is implemented as a custom CUDA/Triton kernel and is not straightforwardly exportable to ONNX, TensorFlow Lite, CoreML, or SNPE, with the official repository reporting that the recursive form of the scan required for inference resists standard `torch.onnx.export` translation. Non-CUDA hardware (AMD GPUs, mobile NPUs, embedded accelerators) requires either custom porting via HIPIFY-style translation or a pure-PyTorch fallback that runs roughly 3–4$\times$ slower than the hardware-aware kernel. Finally, real-world clinical validation of Mamba-based medical models remains limited: most reported results derive from retrospective evaluations on a small number of public datasets, with prospective deployment evidence, external validation, and regulatory submissions yet to materialize. Together, these factors suggest that SSMs should currently be treated as a promising but emerging paradigm rather than a deployment-ready replacement for established architectures.

In summary, Mamba-based SSM architectures represent a qualitative departure from attention-centric design: by replacing quadratic interactions with linear state propagation, SSMs enable global context modeling at a scale previously incompatible with clinical deployment constraints, and while they are not a universal replacement for CNNs or hybrid models, their effectiveness in high-resolution and resource-limited settings positions them as a foundational building block for the next generation of efficient medical imaging systems.

4.4 Comparative analysis and hardware benchmarking

The evolution from CNNs to Mamba represents a continuous search for the optimal Pareto frontier between diagnostic fidelity and computational cost. To provide a holistic view of this landscape, we synthesize the architectural trade-offs of each paradigm in Table 1, a qualitative taxonomy that highlights how successive generations have shifted from rigid inductive biases (CNNs) to selective attention (Hybrids) and finally to linear global context (SSMs), each addressing specific clinical limitations. However,

Table 1 Unified taxonomy of deployment-oriented backbone architectures in medical imaging

Paradigm	Representative models	Core design principle	Key deployment strengths	Intrinsic limitations & Clinical trade-offs
Lightweight CNNs	MobileNet, ShuffleNet, MobileOne, ESPNet, MK-UNet, U-Lite	Strong inductive bias via locality and weight sharing; efficiency through depthwise, grouped, and re-parameterized convolutions	Lowest latency on edge devices Hardware-friendly memory access patterns Mature toolchain support (CPU, GPU, NPU) Proven real-time performance in ultrasound and endoscopy	Limited long-range dependency modeling Global context requires architectural workarounds (dilation, pyramids) Performance degrades in multi-organ or volumetric reasoning tasks
Hybrid CNN–transformer	MobileViT, LeViT, SegFormer, LeViT-UNet, ESFPNet, MedViT, EfficientFormer, iFormer	Selective global context modeling via constrained self-attention embedded within convolutional pipelines	Balanced local efficiency and global awareness Better performance under limited annotations Effective for multi-scale and multi-organ segmentation Achievable real-time inference when attention is restricted	Attention mechanisms remain memory and latency intensive Performance sensitive to resolution and tokenization strategies Harder to optimize across heterogeneous hardware platforms
State space models	VMamba, LightM-UNet, Ultralight VM-UNet, MedMambaLite	Linear-complexity global context propagation via state updates instead of pairwise attention	True global receptive field with $O(N)$ complexity Exceptional parameter and energy efficiency Scales effectively to high-resolution and long sequences Strong fit for battery-powered and Green AI deployment scenarios	Weaker spatial inductive bias compared to CNNs Performance sensitive to scan order and architectural design choices Less mature software and hardware ecosystem

architectural elegance does not guarantee deployment efficiency, as discrepancies between theoretical complexity (FLOPs) and physical latency often arise from hardware-specific constraints, memory access patterns, and compiler maturity.

To ground these architectural trade-offs in deployment reality, Table 2 organizes representative models into two categories: those with *verified mobile latency measurements* and those demonstrating *parameter efficiency without platform-specific benchmarks*. This division reflects a critical methodological constraint in current literature. While parameter counts are architecture-intrinsic properties, latency is a hardware-compiler co-design outcome that requires explicit measurement under controlled conditions. The scarcity of standardized edge-device benchmarks, particularly for recent SSM-based architectures, prevents comprehensive cross-paradigm latency comparison and motivates our focus on models with reproducible deployment evidence.

Among architectures with verified iPhone 12 NPU latency, pure CNNs like MobileOne-S0 establish the performance ceiling in this verified subset at sub-millisecond inference (0.79 ms), while dimension-consistent hybrids like EfficientFormerV2-S0 demonstrate that carefully designed Transformers can approach this efficiency frontier with only 14% overhead (0.9 ms), contrasting sharply with earlier designs like MobileViT-XS that incur 8× slowdown (7.3 ms) due to tensor reshaping operations fragmenting computational graphs. Meanwhile, models like Ultralight VM-UNet achieve remarkable parameter compression (0.049M), yet lack corresponding latency validation on edge devices, reflecting the infrastructural reality that SSM-based selective state-space

Table 2 Quantitative benchmarking of representative deployment-ready architectures

Model	Task	Paradigm	Target hardware	Input Res.	Latency	Params (M)
Verified mobile NPU benchmarks						
MobileOne-S0 [43]	Classif	CNN	iPhone 12 NPU	224×224	**0.79 ms**	2.1
EfficientFormerV2-S0 [51]	Classif	Hybrid	iPhone 12 NPU	224×224	0.9 ms	3.5
MobileViT-XS [50]	Classif	Hybrid	iPhone 12 NPU	256×256	7.3 ms	2.3
Representative lightweight architectures						
ESPNet [48]	Segment	CNN	–	–	100+ FPS†	0.36
LightM-UNet [64]	Segment	SSM	–	–	–	1.0
Ultralight VM-UNet [65]	Segment	SSM	–	–	–	**0.049**
LeViT-UNet-128 [54]	Segment	Hybrid	–	–	114 FPS†	15.9
VM-UNet-B [70]	Segment	SSM	–	–	–	27.4
MedMamba-B [67]	Classif	SSM	–	–	–	∼24

Latency measurements for mobile NPU models use CoreML compilation on iPhone 12 as reported in original publications. †FPS metrics reported without specific hardware context in source papers. Missing values (–) indicate absence of quantitative measurements in original publications or inability to verify experimental conditions. Input resolutions are reported where explicitly stated in source papers; missing resolutions reflect variability in evaluation protocols across studies. Parameter counts reflect architectural properties independent of deployment platform

Metrics are reported from original publications with verified experimental conditions. Latency measurements are hardware- and compiler-specific; direct cross-platform comparisons should account for toolchain differences. Input resolutions also differ across rows (e.g., 224×224 vs. 256×256), and latency typically scales super-linearly with resolution because of the joint growth of tokens and attention cost; consequently, the values in this table are best read as indicative rather than as strict head-to-head comparisons.

Bold values indicate the best-performing model for each metric within the corresponding comparison setting

operators often rely on custom CUDA kernels without native support in mobile inference frameworks such as CoreML, TensorFlow Lite, or Qualcomm SNPE. This sparse latency landscape underscores broader methodological fragmentation in efficient deep learning research: reported metrics vary across batch sizes, warmup protocols, numerical precision, and backend engines, while energy consumption measurements range from unreliable software profiling to rarely documented instrumented power rails, precluding unified efficiency ranking without standardized protocols analogous to MLPerf [71]. Consequently, for immediate clinical deployment on battery-powered devices, only architectures with verified mobile latency represent evidence-based choices, while SSM-based and ultra-lightweight models offer compelling parameter-accuracy trade-offs for server-side inference, though practitioners should benchmark candidates on target hardware rather than relying on reported metrics from potentially incomparable experimental setups. The future viability of SSM architectures at the point of care depends on two critical developments: the maturation of selective scan kernels in mobile inference engines and the establishment of community-wide benchmarking standards that enable reproducible latency claims.

Taken together, the evidence reviewed in this section supports a clear conclusion: no single paradigm dominates across all clinical tasks, modalities, and hardware targets. Lightweight CNNs remain the most reliable choice when sub-millisecond latency, mature toolchain support, and proven hardware compatibility are paramount, particularly for interventional applications such as endoscopy or ultrasound on mobile-class devices. Hybrid CNN–Transformer architectures occupy a middle ground that is most attractive when global context is required, when annotated data are limited, and when target hardware supports efficient attention execution at moderate resolutions. State Space Models offer the strongest theoretical scaling and the lowest parameter counts in the current literature, making them the most promising option for high-resolution 2D and volumetric tasks where attention becomes prohibitive, but their advantages are

partly offset by software immaturity, scan-order sensitivity, and limited prospective validation. The appropriate paradigm is therefore best determined by jointly considering modality, input resolution, clinical latency budget, and the maturity of the target inference stack, rather than by accuracy benchmarks alone. This task- and hardware-conditioned view motivates the modality-aware decision pathway summarized in §7.

5 Balancing efficiency and fidelity: model compression in medical AI

The deployment gap between state-of-the-art deep learning models and real-world clinical environments is often not architectural, but computational in nature. Even when efficient backbones such as lightweight CNNs, hybrid Transformers, or Mamba are employed, the resulting models frequently exceed the memory, latency, or energy budgets of PoC devices. Consequently, model compression must be viewed not merely as a post hoc optimization, but as a foundational requirement for the democratization of medical AI [72–74].

To address these constraints, contemporary literature converges on four orthogonal but complementary paradigms: network pruning, quantization, knowledge distillation, and low-rank factorization. Table 3 provides a high-level comparison of these strategies, summarizing their core mechanisms and the critical trade-offs between computational efficiency and diagnostic safety. In the following sections, we examine the specific implementation and clinical implications of each approach.

5.1 Network pruning

Network pruning reduces computational complexity by eliminating parameters, channels, or tokens that contribute minimally to the inference process. In medical imaging, pruning is particularly appealing due to the inherent sparsity of clinically relevant information. Large portions of scans often contain background, air, or healthy tissue.

Table 3 Comparison of model compression strategies for deployable medical AI

Method	Core mechanism	Key advantages	Limitations and risks
Network pruning	Eliminates redundant parameters, channels, or tokens	Significant FLOP and memory reduction Aligned with spatial sparsity in medical images Enables dynamic, region-aware inference	Degrades sensitivity to rare or long-tailed pathologies Loss may not appear in aggregate metrics Requires safety-aware evaluation
Quantization	Reduces numerical precision of weights and activations	Direct acceleration on edge NPUs Major model-size and energy reductions Often deployable post-training	Attenuates fine-grained signal representations Sensitive to distributional shift Hardware- and platform-dependent
Knowledge distillation	Transfers decision knowledge from teacher to student model	Preserves clinically meaningful boundaries Flexible across architectures and tasks Supports safety-aware compression	Depends strongly on teacher quality and calibration Hyperparameter-sensitive training Knowledge loss under teacher–student mismatch
Low-rank factorization	Approximates operators using low-rank matrix products	Principled reduction of algebraic redundancy Implicit regularizer for small datasets Well-suited to volumetric data	Effectiveness collapses when representations are not actually low-rank Risk of oversimplification on heterogeneous pathologies Requires task-specific rank selection

A representative example is APFormer, proposed by Lin et al., which introduces adaptive token pruning into Transformer-based segmentation [75]. Unlike uniform attention mechanisms that process all patches equally, APFormer dynamically discards redundant tokens corresponding to non-informative regions during inference. This region-aware pruning achieves substantial GFLOP reductions while preserving segmentation accuracy, illustrating how task structure can guide safe compression.

However, pruning introduces unique risks in medical contexts. Holste et al. demonstrated that on long-tailed datasets, where rare pathologies are underrepresented, pruning can disproportionately impair the detection of minority classes, even when aggregate metrics such as AUC remain unchanged [76]. These "pruning-identified exemplars" (PIEs) expose a critical limitation: aggressive sparsification may silently erode sensitivity to rare but clinically significant findings. Consequently, pruning strategies must be evaluated beyond global performance metrics, with explicit attention to safety-critical edge cases.

5.2 Quantization

Quantization reduces model size and accelerates inference by lowering numerical precision, typically converting FP32 representations to INT8, FP16, or even ternary formats. For edge devices equipped with NPUs, quantization is often the most direct pathway to real-time performance.

Early studies demonstrated that post-training INT8 quantization enables up to 97% latency reduction for cancer diagnosis models deployed on portable devices, transforming offline pipelines into real-time clinical tools [77]. Subsequent work identified 8-bit precision as a practical compromise, achieving 4–8$\times$ parameter reduction while retaining over 80% diagnostic accuracy [78].

Nevertheless, quantization is not without trade-offs. Reducing numerical precision may attenuate fine-grained signal representations, which can be critical in medical images where diagnostically relevant patterns are often subtle. Quantized models may also exhibit increased sensitivity to data distribution shifts, particularly when deployment conditions differ from those used for calibration. Additionally, very low-bit implementations often rely on hardware-specific support, which can limit cross-platform generalizability and complicate large-scale clinical deployment.

A practical complication rarely discussed in primary studies is hardware heterogeneity across mobile NPUs. Clinically motivated mixed-precision schemes, for example, keeping the input stem and classification head in FP16 to preserve subtle signal representations while quantizing intermediate blocks to INT8, require native FP16 support on the target accelerator, which many widely deployed devices lack. The Qualcomm Hexagon DSP, common in field-deployed Snapdragon devices, historically lacks FP16 units in its vector co-processor and supports only per-tensor or per-channel integer quantization in its production software stack; on such platforms, mixed-precision schemes fall back to slower CPU execution or fail to compile. Recent flagship silicon has narrowed this gap with INT4/INT8/INT16/FP16 mixed precision, but this is not yet typical in the older or mid-range devices that dominate point-of-care settings. Practically, pure INT8 with quantization-aware training is the most portable choice for heterogeneous mobile fleets, while any mixed-precision deployment should explicitly state its hardware target, as the

same model may achieve real-time latency on a flagship NPU and fail to load on a two-generation-older device.

5.3 Knowledge distillation

Knowledge Distillation (KD) compresses models by transferring knowledge from a large, well-trained teacher model to a compact student model. Unlike pruning or quantization, KD does not remove capacity directly; instead, it reshapes the learning objective so that lightweight models inherit clinically meaningful decision boundaries.

Alabbasy et al. demonstrated that distilled students deployed via TensorFlow Lite can retain diagnostic accuracy within 1–2% of heavyweight teachers, validating KD as a deployment-ready strategy [79]. More recent work tailors student architectures to specific clinical domains. ConvMixer-based students have been designed for pulmonary disease detection, while global context from Vision Transformer teachers has been distilled into convolutional U-Net students, effectively embedding long-range anatomical reasoning into efficient CNNs [80, 81].

Advanced distillation schemes address specific compression pathologies. Rank-Sensitive Knowledge Distillation (RSKD) preserves attention hierarchy during compression, maintaining global contextual understanding despite reduced model capacity [82]. Similarly, KD-enabled denoising frameworks reconstruct diagnostically reliable images on resource-constrained edge devices [83].

However, KD's suitability for medical imaging introduces critical dependencies and failure modes that warrant careful consideration. The student model inherits not only the teacher's capabilities but also its systematic biases and calibration errors, a particularly concerning issue in medical contexts where silent error propagation can compromise diagnostic safety without triggering obvious performance degradation during validation. The distillation process itself introduces substantial methodological complexity through hyperparameter sensitivity across temperature scaling, loss weighting, and intermediate supervision layers, making reproducible cross-task deployment difficult and limiting the generalizability of distillation protocols. Furthermore, architectural heterogeneity between teachers and students creates a fundamental bottleneck in knowledge transfer. When student capacity or inductive biases differ significantly from the teacher, certain representational structures cannot be faithfully compressed, potentially losing clinically important feature interactions that lack explicit supervision signals in standard distillation objectives. This structural mismatch becomes especially problematic in medical imaging, where subtle spatial relationships and rare pathological patterns may reside precisely in those hard-to-transfer representational spaces that conventional distillation losses fail to capture.

A specific and clinically consequential variant of this concern is whether distillation transfers probability calibration. Standard KD objectives align soft predictions via KL divergence, but the temperature is typically tuned for accuracy rather than calibration fidelity, and students often inherit the teacher's overconfidence rather than its calibration [84]. Calibration-aware variants (e.g., temperature-scaled teachers, spherical logit matching, or KD combined with explicit calibration losses) have been proposed in the general computer-vision literature [85] but remain rarely applied in medical-imaging KD studies. In safety-critical settings, we recommend that distilled medical models report

Expected Calibration Error (ECE) and reliability diagrams before and after distillation, and consider post-hoc recalibration of the student as a minimum baseline.

5.4 Low-rank factorization

Low-rank factorization approximates large convolutional or attention matrices using products of smaller matrices, reducing both parameter count and computational complexity. Beyond compression, this technique acts as an implicit regularizer, which is advantageous for medical datasets that are often limited in size.

Low-rank constraints have been applied to 3D convolutional kernels in volumetric brain tumor segmentation, substantially reducing model size while maintaining competitive Dice scores [86]. By explicitly modeling redundancy in high-dimensional medical features, low-rank factorization achieves compression without relying on post hoc sparsification, offering a principled alternative for volumetric imaging tasks.

However, the benefits of low-rank factorization are contingent on the extent to which redundancy is truly present in the underlying feature representations. When pathological patterns are spatially localized or highly heterogeneous, aggressive rank constraints may oversimplify the feature space, leading to a loss of discriminative capacity. In practice, selecting an appropriate rank often requires empirical tuning and task-specific insight, which can limit the method's robustness and transferability across diverse medical imaging modalities.

5.5 Synthesis: clinical fidelity after compression

Across the four compression paradigms, the primary literature has converged on broadly consistent reporting conventions, though gaps remain in how compression-induced degradation is characterized. Quantitatively, an accuracy or Dice drop of one to two percentage points relative to the uncompressed baseline is the de facto threshold most studies treat as clinically acceptable when reporting headline metrics [79], with 8-bit post-training quantization typically retaining over 80% of full-precision diagnostic performance [78] and well-tuned knowledge distillation closing the gap to within 1–2% of the teacher [79]. Beyond aggregate metrics, however, four failure modes are inconsistently reported and warrant explicit attention. First, calibration drift: compressed models, particularly those subject to aggressive quantization or distillation, often retain their decision boundaries but become overconfident or underconfident in their probability estimates, a property that aggregate AUC and Dice fail to capture [76]. Second, rare-class and long-tailed sensitivity: aggressive pruning has been shown to disproportionately erode performance on minority pathologies even when global metrics remain unchanged, exposing a silent failure mode of direct clinical consequence [76]. Third, subgroup robustness: bias inherited from the teacher, the calibration set, or the original training distribution can be amplified rather than revealed by compression, with disparate impact across demographic or acquisition-protocol subgroups that is rarely audited in the deployment literature. Fourth, distributional shift: post-training compression is typically validated under the same acquisition conditions used for calibration, with limited evaluation against scanner, protocol, or institutional shifts that are routine in clinical practice. We therefore recommend that future deployment-oriented studies report at minimum (i) the compression-induced change in primary metric with confidence intervals, (ii) Expected Calibration Error (ECE) before and after compression, (iii) per-class

or per-subgroup sensitivity for clinically significant minority classes, and (iv) at least one external or shifted-distribution evaluation. These reporting elements are reflected in Table 4 and operationalize the broader principle that, in safety-critical settings, compression must be evaluated as a fidelity-preserving transformation rather than a purely computational optimization.

6 Discussion

6.1 Navigating the architectural landscape: mamba, hybrids, and the 3D bottleneck

Rather than exhaustively cataloging architectural variants, this discussion focuses on the practical implications of recent design choices for deployment-oriented medical imaging systems. A central dilemma for practitioners in the current era is selecting an appropriate backbone amid the growing divergence between attention-based and state-space architectures. Increasingly, this decision is shaped not only by accuracy, but by the alignment between a model's inductive bias and the spatial structure of the target modality.

For high-resolution 2D modalities such as whole-slide pathology or chest X-ray, the linear complexity of Mamba-based models offers a compelling advantage under memory-constrained settings. However, empirical studies suggest that the choice of scan strategy plays a critical role in translating this theoretical efficiency into practice. While wavefront-style scans are conceptually appealing and biologically motivated, emerging evidence across recent works indicates that cross-scan strategies tend to provide a more stable balance between long-range context aggregation and memory access efficiency on contemporary hardware. These observations highlight scan order as a task-dependent and modality-dependent design choice rather than a universally optimal solution.

In 3D volumetric imaging tasks such as CT and MRI, the trade-offs become more nuanced. Here, deployment feasibility is often governed by memory bandwidth rather than raw compute. Under memory-limited regimes (e.g., GPUs with less than 8GB VRAM), pure or predominantly SSM-based designs, such as lightweight Mamba-UNet variants, appear better suited to processing larger volumetric patches without incurring

Table 4 Proposed deployment-first reporting checklist for efficiency-oriented medical imaging models.

Category	Recommended disclosure items
Task	Clinical task (classification, segmentation, triage), modality (CT, MRI, US, WSI), input resolution, preprocessing pipeline, dataset scale, class imbalance
Hardware platform	Device model (CPU/GPU/NPU/FPGA), memory capacity, power mode (sustained vs burst), thermal constraints, system load conditions
Software stack	Runtime engine (TensorRT, ONNX Runtime, OpenVINO, CoreML, TFLite), version numbers, backend configuration, compiler optimizations
Numerical precision	Inference precision (FP32/FP16/INT8), quantization method (PTQ vs QAT), calibration dataset size, per-tensor vs per-channel quantization
Inference protocol	Batch size (recommended = 1 for edge), warm-up runs, timed runs, synchronization method, I/O handling scope
Latency reporting	Median and tail metrics (p50, p90), variance across runs, notes on throttling or performance drift
Memory footprint	Model size on disk, peak runtime memory, additional runtime/compiler buffers
Energy & power	Measurement method (on-device sensors or external), idle power subtraction, joules per inference, variance. Distinguish chip-level (e.g., GPU/NPU package) from system-level power (memory, peripherals, display) where battery life is the binding constraint.
Reliability & safety	Post-optimization calibration (ECE, NLL), rare-class sensitivity, robustness to noise/resolution shifts, failure mode analysis

Emphasizes contextual disclosure rather than prescriptive benchmarking, enabling interpretable comparison across heterogeneous edge environments

the quadratic memory growth associated with full attention mechanisms. Conversely, for diagnostic tasks that hinge on subtle local texture cues, such as detecting early ground-glass opacities, hybrid Transformer architectures may remain a more conservative choice, leveraging the well-established representational strengths of convolutional stems combined with selective attention, provided hardware resources permit. However, the empirical evidence for SSM-based architectures in 3D volumetric medical imaging remains limited: most reported results derive from a small number of retrospective benchmarks, often on single-institution datasets and without external or prospective validation. Conclusions about SSM superiority in volumetric settings should therefore be regarded as emerging rather than established, and direct comparisons with mature hybrid baselines under matched hardware and training budgets are still needed.

A promising emerging direction is the "asymmetric hybrid" paradigm, in which efficient 2D SSM encoders are used for slice-wise feature extraction, followed by lightweight 3D aggregation to enforce volumetric consistency. Such designs respect the anisotropic nature of many medical volumes while offering a pragmatic compromise between accuracy, throughput, and memory consumption. Beyond the alignment with anisotropic acquisition geometry, this paradigm offers three concrete advantages relevant to deployment. First, it substantially reduces the parameter and memory footprint of the 3D stage: by performing the most representationally demanding computation in 2D, the volumetric component can be confined to lightweight feature aggregation rather than full 3D convolution or attention, sidestepping the cubic growth in compute and memory that pure 3D backbones incur. Second, it improves robustness to inter-slice variability common in CT and MRI acquisition, including non-uniform slice thickness, partial-volume artefacts, and motion-induced misregistration, since the 2D encoder operates independently per slice and the aggregation stage learns to reconcile rather than memorize slice-level inconsistencies. Third, it enables practical reuse of pre-trained 2D backbones, which substantially eases data-scarcity constraints in 3D medical imaging where large annotated volumetric datasets remain rare. These properties make the asymmetric hybrid a natural fit for edge-deployable volumetric inference, though formal benchmarking against fully 3D SSM and Transformer baselines remains an open question.

6.2 The "Compiler Gap": bridging theory and real-world latency

Despite rapid architectural innovation, a persistent gap remains between theoretical efficiency and realized deployment performance. Our review highlights that commonly reported metrics such as FLOPs and parameter counts often fail to predict end-to-end latency or energy consumption in clinical systems. This discrepancy stems largely from the uneven maturity of software toolchains supporting different operator families.

Convolutional networks currently enjoy a decisive deployment advantage, benefiting from decades of optimization in mature ecosystems such as NVIDIA TensorRT, OpenVINO, and CoreML, where depthwise and grouped convolutions are routinely fused and scheduled for efficient execution. In contrast, many Mamba-based and SS2D-based models rely on custom kernels that lack native support in widely used inference engines, including ONNX Runtime and TFLite Micro. As a result, architectures that are mathematically efficient may exhibit suboptimal performance on heterogeneous or non-NVIDIA edge platforms, such as mobile DSPs or low-power CPUs.

This "compiler gap" has direct clinical implications: unpredictable latency and energy behavior can undermine real-time decision support and disrupt workflow integration at the point of care. In the near term, highly optimized CNN-based models therefore remain the most reliable choice for large-scale deployment across diverse hardware fleets. Closing this gap for SSM-based architectures will require a coordinated co-design effort spanning model architecture, operator standardization, and compiler development, potentially leveraging emerging frameworks such as TVM or Triton to bridge research prototypes and production systems.

Beyond standardization, several compiler-level techniques offer a purely software-based path to narrowing this gap without requiring hardware vendors to ship new operators. Tensor-compilation frameworks such as TVM [35] and TensorRT-LLM apply operator fusion, layout transformation, and auto-scheduling to compose efficient kernels from primitive operations, with the original TVM evaluation reporting $1.2\times$ to $2\times$ speedups from fusion alone on conventional CNN and RNN workloads. Triton-style domain-specific languages further allow researchers to express the parallel-scan recurrence at a level above hand-written CUDA, which both improves portability and exposes the operation to graph-level compiler analysis. Concrete proofs of feasibility already exist: hardware-aware Mamba implementations have been demonstrated on AMD ROCm via HIPIFY translation and on AWS Neuron via custom NKI kernels, with the latter reporting nearly three orders of magnitude speedup over the reference PyTorch fallback. These results suggest that the compiler gap for SSMs is not a fundamental obstacle but a software-engineering one, and that targeted investment in fused selective-scan templates within mainstream inference compilers (ONNX Runtime, OpenVINO, TVM) could substantially extend the deployable surface of Mamba-class architectures. The remaining limitation is coverage rather than capability: such optimizations currently require backend-specific manual work, and no mainstream compiler yet performs end-to-end auto-fusion of selective-scan operators across heterogeneous mobile and embedded targets.

6.3 Safety-critical optimization: quantization and calibration

Before discussing optimization-induced safety risks, it is worth acknowledging that several data-level challenges precede and compound them. First, dataset bias is pervasive in medical imaging benchmarks: training cohorts are often skewed toward specific scanner vendors, acquisition protocols, demographic groups, and tertiary-care institutions, and compression has been shown to amplify rather than reveal such bias by reducing the model's capacity to learn minority-group features. Second, annotation quality is uneven: inter-rater disagreement is well documented in segmentation tasks (Dice variability of 5–15% between expert annotators is not uncommon for soft-tissue boundaries), and lightweight student models trained on noisy labels are less able to absorb this noise than larger teachers, making distilled or pruned models more sensitive to annotation inconsistency. Third, domain shifts across institutions routinely degrade models trained at one site when deployed at another, and this shift is rarely characterized in the same study that proposes a new efficient architecture. These three challenges are not solved by compression or by efficient backbones, but they decisively shape whether an efficient model remains safe when moved from the benchmark into a clinical setting. We accordingly recommend that deployment-oriented studies report at least one external-cohort

or shifted-distribution evaluation, document the demographic and acquisition composition of their training data, and treat inter-rater label noise as an explicit source of model uncertainty.

As medical AI systems are increasingly pushed toward edge deployment, aggressive optimization techniques such as quantization become difficult to avoid. In healthcare settings, however, optimization must be framed not merely as an engineering problem, but as a safety-critical design decision. Reducing the numerical precision from FP32 to INT8 can substantially improve latency and energy efficiency, yet it can also degrade uncertainty calibration or disproportionately affect sensitivity to rare but clinically significant findings.

Importantly, a quantized model may retain strong aggregate performance metrics, such as global Dice or AUC, while becoming subtly overconfident in incorrect predictions or less responsive to long-tail pathologies. To address this risk, we argue that the evaluation of deployment-ready models should routinely extend beyond accuracy to include safety-oriented indicators such as Expected Calibration Error (ECE), Negative Log Likelihood (NLL), and rare-class sensitivity, particularly in post-quantization settings.

Across the reviewed literature, Quantization-Aware Training consistently emerges as more reliable than Post-Training Quantization for medical imaging tasks, as it allows models to adapt their decision boundaries to reduced-precision representations. In addition, mixed-precision strategies, preserving higher precision in the first and last layers while aggressively quantizing intermediate blocks, offer a pragmatic compromise that maintains diagnostic fidelity while delivering meaningful efficiency gains. Reporting such design choices and their calibration impact should be considered best practice rather than optional.

6.4 The imperative of explainability in compact models

As models become smaller and more aggressively optimized, the role of explainability shifts from a supplementary feature to a core component of validation. Lightweight architectures produced through pruning, distillation, or quantization risk achieving efficiency by exploiting dataset-specific shortcuts rather than learning clinically meaningful representations. In this context, explainability mechanisms play a crucial role in verifying that compression has not eroded the causal basis of model predictions.

For edge-deployed systems operating with limited or no human-in-the-loop oversight, explainable AI serves as an important safety valve. Post hoc techniques such as saliency mapping or attention visualization can support auditing and error analysis, helping clinicians assess whether automated decisions align with anatomical or pathological expectations. As a concrete illustration of such erosion, Grad-CAM has been shown to expose shortcut learning in compressed chest-radiograph triage models, where aggressively pruned or distilled networks shift attention from clinically relevant regions (lung parenchyma, nodule margins) to acquisition artifacts such as chest tubes, ECG leads, or text annotations, a failure mode that can leave the headline AUC essentially unchanged while invalidating the diagnostic basis of prediction. Perturbation-based methods such as LIME and SHAP can reveal similar erosions by occluding image regions and observing prediction shifts, providing complementary, higher-granularity evidence. However, these techniques exhibit a clear interpretability versus efficiency trade-off that is

especially consequential at the edge. Class-activation methods such as Grad-CAM require only a single forward and backward pass, adding a few percent to inference time and remaining compatible with real-time deployment. Path-integrated gradient methods such as Integrated Gradients are mathematically more rigorous but approximate the path integral over a baseline using 20–300 interpolation steps, each demanding a forward and backward pass, placing them well above Grad-CAM in cost and impractical for streaming or per-frame use on edge hardware. Perturbation-based methods (LIME, SHAP) typically require hundreds to thousands of forward passes per explanation and are usually infeasible on battery-powered devices. Attention-based visualization is informative for Transformer and hybrid models at near-zero additional cost, but transfers poorly to SSM-based architectures, where the absence of explicit attention maps leaves a tooling gap that has yet to be filled. More broadly, these considerations motivate growing interest in intrinsic explainability, designing efficient architectures whose internal representations are interpretable by construction, rather than relying exclusively on post hoc analysis to justify opaque models.

6.5 Synergy between efficiency, privacy, and equity

The implications of architectural efficiency extend beyond individual device performance to influence the broader ecosystem of medical AI deployment. There is a natural synergy between lightweight models and privacy-preserving paradigms such as federated or split learning. Federated learning has matured into a deployable framework for medical imaging, with established surveys documenting its clinical adoption patterns and infrastructure requirements [87], and active research addressing practical issues such as client-side heterogeneity in representation alignment when models are trained across institutions with diverse acquisition protocols [88]. In collaborative training settings, communication bandwidth often represents the dominant bottleneck; here, compact architectures, low-rank adaptation, and quantized updates can substantially reduce synchronization costs, enabling participation by institutions with limited infrastructure.

The energy dimension of Green AI ties directly to specific architectural and compression choices documented earlier in this review. Architecturally, structural re-parameterization (e.g., MobileOne) and depth-wise separable convolutions reduce inference energy by collapsing multi-branch training graphs into single-path inference paths that fuse cleanly on mobile NPUs, while parallel Vision Mamba layers in Ultralight VM-UNet achieve segmentation with 0.06 GFLOPs and a corresponding reduction in active compute energy. On the compression side, INT8 quantization with quantization-aware training typically reduces both model footprint and per-inference energy by an order of magnitude over FP32 baselines on integer-friendly accelerators, and hardware-aware distillation has been reported to improve energy efficiency by 63% on NVIDIA Jetson Orin Nano in the case of MedMambaLite relative to its non-optimized teacher. These figures illustrate that the energy savings highlighted in Fig. 1 are not abstract: they accrue from concrete, stackable design decisions at the architecture and compression layers, and they translate directly into longer battery life for point-of-care devices and reduced operational cost for institutionally deployed inference servers.

Efficiency also plays a central role in promoting equity. By enabling advanced diagnostic models to operate on battery-powered or low-cost hardware, efficient AI systems can help extend imaging-based decision support to resource-constrained and rural

environments. From this perspective, the shift toward compact and energy-efficient architectures is not merely a technical optimization, but a foundational step toward more sustainable, accessible, and inclusive digital health systems.

6.6 Toward a deployment-first evaluation standard for medical imaging models

As efficiency-driven architectures increasingly target edge and point-of-care deployment, the field faces a critical tension between the demand for quantitative evidence and the absence of standardized evaluation practices. While recent studies routinely report latency, memory footprint, or energy metrics, these numbers often emerge from heterogeneous and poorly specified conditions. Variations in hardware platforms, runtime engines, numerical precision, batch sizes, and warm-up protocols can induce order-of-magnitude performance differences. Without explicit disclosure, such variability risks being misattributed to architectural superiority rather than recognized as an artifact of evaluation setup.

This challenge is particularly acute in medical imaging, where deployment constraints are operational rather than abstract. A model labeled "efficient" under GPU-centric benchmarking may fail real-time requirements on CPU-only workstations, mobile NPUs, or embedded accelerators. Conversely, architectures optimized for specific runtimes may appear disproportionately strong when evaluated with favorable toolchains. Absent a common reporting baseline, cross-study comparison becomes unreliable, leaving clinicians and system designers without context to assess whether reported efficiency gains transfer to their deployment environments.

Commonly reported proxies (FLOPs, parameter count, or theoretical GOPS/J) provide incomplete views of deployability. FLOPs ignore memory access patterns and operator fusion; GOPS/J is rarely measured under controlled power instrumentation and conflates architectural properties with hardware-specific characteristics. When reported in isolation, these metrics provide partial and sometimes misleading characterizations of real-world performance.

For deployment-oriented reporting we therefore advocate a minimum quantitative set that captures the four dimensions most consequential for clinical use: (i) Memory Access Cost (MAC), since on edge accelerators latency is increasingly bound by memory bandwidth rather than arithmetic intensity; (ii) energy per inference under a stated power-measurement convention, distinguishing chip-level from system-level draw as discussed above; (iii) calibration fidelity (Expected Calibration Error and reliability under post-compression conditions), to surface silent reliability degradation invisible to AUC or Dice; and (iv) per-class or rare-class sensitivity, since aggregate accuracy can mask disproportionate failures on minority pathologies. Table 4 operationalizes this set as part of the proposed reporting checklist.

Optimization techniques further complicate evaluation. Quantization, pruning, and operator re-parameterization can dramatically reduce latency and energy consumption while potentially degrading calibration, rare-class sensitivity, or numerical stability, clinically meaningful properties seldom reported alongside efficiency gains. Two models with identical latency profiles may differ substantially in diagnostic reliability, a distinction invisible under conventional benchmarking.

Rather than proposing another centralized benchmark, we advocate for a deployment-first reporting standard: minimal but explicit disclosure of conditions under which

efficiency claims are obtained. This standard does not mandate uniform hardware or datasets but makes efficiency claims interpretable by anchoring them to clearly defined hardware–software stacks, inference protocols, and measurement procedures.

Table 4 presents a proposed checklist for reporting efficiency-oriented results. This framework is intentionally lightweight, not a prescriptive benchmark protocol requiring multi-device evaluation, but a set of contextual disclosures enabling readers to situate metrics within their operational scope. By making assumptions explicit (task definition, runtime engines, precision, measurement procedures), authors reduce ambiguity while preserving flexibility. Importantly, this approach acknowledges uncertainty as intrinsic to deployment evaluation. Reporting median and tail latency (e.g., p90), warm-up behavior, and cross-run variance recognizes that edge performance is non-deterministic, particularly on shared or thermally constrained devices. Similarly, documenting calibration strategies and post-quantization reliability metrics reframes efficiency as a safety-critical property, not merely a computational achievement. In addition, energy reporting should distinguish between chip-level power (e.g., accelerator package draw via on-die sensors or `nvidia-smi`) and system-level power (which additionally captures memory, peripherals, display, and idle losses); the latter is typically what limits battery life on portable medical devices, and chip-level measurements alone can substantially underestimate operational energy budgets.

The benchmark presented below is intended as an illustrative demonstration of the proposed reporting checklist for two representative architectures, not as a definitive head-to-head ranking of efficient model families. Observed orderings depend jointly on the runtime engine, numerical precision, batch size, and target processor and may shift under different configurations; the readers should accordingly treat the comparison as a working example of disclosure rather than a generalizable performance verdict.

To demonstrate this framework's utility, we provide a reference implementation benchmarking two representative efficient architectures, MobileOne-S0 (pure CNN) and EfficientFormerV2-S0 (Hybrid), under standardized deployment assumptions. Table 5 documents the exact evaluation context, while Table 6 reports the measured performance metrics across both CPU and GPU substrates.

These results, grounded in the declared context, reveal critical deployment dynamics. First, despite higher thermal design power, GPU inference validates the "race-to-idle" principle by consuming less net energy (0.60 J) than the CPU (0.94 J) due to its faster execution, thereby minimizing the total power integral. Second, architectural affinity plays a measurable role: under the evaluated conditions, the pure CNN (MobileOne) achieves a significantly higher GPU acceleration factor ($4.3\times$) than the hybrid EfficientFormerV2 ($2.7\times$), suggesting that dense convolutions leverage massive parallelism more effectively than attention mechanisms at a batch size of one.

Crucially, these performance rankings are strictly bound by the specific hardware–software stack. Altering runtime engines (e.g., TensorRT), quantization schemes, or target processors could fundamentally shift the observed ordering. Ultimately, this standardized disclosure prioritizes reproducibility over absolute rankings, enabling readers to distinguish fundamental architectural properties from environment-specific optimizations. This review has traced the evolution of deployment-ready architectures for medical imaging across three waves: efficient CNNs, hybrid CNN-Transformers, and linear-complexity state space models; and synthesized the model-compression

Table 5 Benchmark configuration and measurement environment.

Category	Parameter	Specification
Target task	Task type	Image classification (ImageNet-1k)
	Input resolution	$224 \times 224 \times 3$
	Batch size	1 (Simulating real-time edge deployment)
Hardware	CPU	Intel Core i7-12700K (12 Cores, 3.6-—5.0 GHz)
	System memory	32GB DDR4-3200
	GPU accelerator	NVIDIA GeForce RTX 3070 Ti (8GB GDDR6X)
Software stack	OS	Linux x86_64 (Ubuntu 22.04 LTS)
	Python environment	Python 3.12
	DL framework	PyTorch 2.x (CUDA 12.x backend for GPU)
Instrumentation	Latency timing	`torch.cuda.Event` (GPU)/`time.perf_counter` (CPU)
	Energy profiling	`perf stat -e power/energy-pkg/` (CPU Package)
		`nvidia-smi dmon` (GPU Power Draw)
Protocol	Precision	FP32 (Single precision)
	Sampling	50 Warm-up runs + 200 Measured runs

Full disclosure of hardware, software, and measurement protocols for the illustrative benchmark presented in §6.6. Abbreviations: *DDR4* = Double Data Rate 4 system memory; *GDDR6X* = Graphics Double Data Rate 6X video memory; *RTX* = NVIDIA Ray Tracing GPU family; *CUDA* = Compute Unified Device Architecture; *RSS* = Resident Set Size (peak runtime memory). The benchmark targets a single-image batch (batch size = 1) to simulate real-time edge inference, where requests typically arrive serially rather than in batches; warm-up and timed-run separation follows standard practice for stable latency measurement on shared, thermally constrained hardware. For practitioners, this configuration represents a desktop workstation rather than a battery-powered point-of-care device, and the GPU here serves as an approximation of a server-side accelerator rather than of a mobile NPU

Table 6 Deployment-oriented efficiency metrics

Model	Params (M)	Median latency (ms)	P90 latency (ms)	Throughput (img/s)	Energy (J/inf)	Peak RSS (MB)
Scenario A: CPU Inference (Intel i7-12700K)						
MobileOne-S0	2.08	5.18	5.36	193	0.94	832
EfficientFormerV2-S0	3.60	11.70	11.94	85	1.64	901
Scenario B: GPU Inference (RTX 3070 Ti)						
MobileOne-S0	2.08	**1.19** ($4.3\times$)	1.23	**842**	**0.60**	1156
EfficientFormerV2-S0	3.60	4.34 ($2.7\times$)	4.44	230	0.67	1267

Measured performance of two representative architectures (a pure CNN, MobileOne-S0; and a hybrid CNN-Transformer, EfficientFormerV2-S0) on commodity CPU versus accelerated GPU substrates, under the conditions disclosed in Table 5. Column definitions: *Params (M)* = trainable parameter count in millions; *Median Latency* = median wall-clock inference time per image across 200 measured runs (p50); *P90 Latency* = 90th-percentile latency, reflecting tail behavior under thermal and scheduling variability; *Throughput* = sustained images processed per second; *Energy (J/inf)* = joules of accelerator-package energy consumed per inference, derived from `perf stat` for CPU and `nvidia-smi` for GPU (chip-level; not system-level); *Peak RSS (MB)* = peak Resident Set Size, indicating maximum host memory occupied during inference. Speedup factors in parentheses are relative to the same model's CPU baseline. The GPU values illustrate the *race-to-idle* principle: faster execution reduces total energy despite higher peak power. Practitioners should note that these rankings are bound to this specific hardware-software stack and may shift under different runtimes (e.g., TensorRT), precisions (INT8 vs. FP32), or target processors (mobile NPU vs. desktop GPU)

Bold values indicate the best-performing model for each metric within the corresponding inference scenario

and reporting practices that determine whether algorithmic efficiency translates into clinical impact. The synthesis supports a unified, modality-aware decision pathway for practitioners: lightweight CNNs remain the most reliable choice for sub-millisecond, latency-bound interventional applications such as ultrasound or endoscopy; hybrid CNN-Transformer architectures are best suited to mid-resolution dense-prediction tasks where global context is required and annotated data are limited; state space models are the most promising option for high-resolution 2D and volumetric inputs where attention becomes prohibitive, with the caveat that their software ecosystem and prospective validation are still maturing; and asymmetric 2D-encoder/3D-aggregator hybrids offer a pragmatic compromise for edge-deployable volumetric inference.

Translation into routine clinical use will further require external and prospective validation across institutions, regulatory clearance (e.g., Food and Drug Administration (FDA), European Conformity – Medical Device Regulation (CE-MDR)), workflow integration with existing Picture Archiving and Communication System (PACS) and Digital Imaging and Communications in Medicine (DICOM) pipelines, and continued attention to calibration, rare-class sensitivity, and subgroup robustness under acquisition and demographic shift. Looking further ahead, advances in compiler co-design, neuromorphic and analog accelerators, and standardized selective-scan kernels may extend the deployable surface of efficient medical AI well beyond current edge constraints. By integrating architectural innovation, system-level efficiency, and clinical trust, future lightweight models can move beyond benchmark performance and support sustainable, equitable, and trustworthy medical AI deployment.

Author contributions
T.S.H. conceived the study and supervised the project. C.M.N. conducted the literature review and wrote the majority of the manuscript. T.S.H. provided guidance on the scope and structure of the review and edited the manuscript. All authors reviewed and approved the final manuscript.

Data availability
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Declarations

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